Muhammad Zaryab Safder

AI and software engineer | BSc Computer Science (2026)
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PROFILE

Computer science graduate who builds AI systems end to end (web and mobile front ends, backend, deployment). Co-built a camera-based fall detection platform as an industrial research and development final-year project; now an AI engineer on LLM-based tools. Seeking research opportunities.

Research interests: software engineering for AI, LLMs for code, empirical software engineering, reliability of AI systems, cyber security (software supply chain, malware detection).

EDUCATION

BSc Computer Science, COMSATS University Islamabad, Lahore Campus | 09/2022 to 07/2026
CGPA 3.08/4.00. Final-year project graded A. Coursework: Machine Learning, Artificial Intelligence, Computer Vision, Parallel and Distributed Computing, Operating Systems, Algorithms, Databases, Linear Algebra, Statistics.

PROJECTS

Eldercare: patient monitoring and fall detection platform | final-year project, industrial research and development, two-person team

- **My part, the system around the model:** Flutter web and Android app, FastAPI backend, self-hosted PostgreSQL and Redis, alerts through WebSockets and Firebase push notifications, Docker deployment, integration testing, and most of the project report.
- **Team result:** the detection model, MAGnet (a multimodal attention graph network, built and trained by my teammate), reached 98.2% accuracy and 99.78% ROC AUC on 38,674 windows from two public datasets (NTU RGB+D 120, FallVision). Falls only; seizure detection is planned future work.
- Co-author of a research manuscript based on the project (in final stages of writing).

Empirical software engineering projects | github.com/Zaryab111

- **bugfix-tests-mining**: mined 12,537 commits from 15 Python projects to ask whether bug fixes come with tests. The fix-keyword rule was 30% precise on a labelled sample; the first result (50.0% vs 46.2%) did not survive a fair comparison (41.7% vs 38.8%, p = 0.059).
- **rest-api-lint**: checker for REST API design rules, run on 200 public OpenAPI specifications. Precision was 39% on 100 flagged issues checked against a written rule; after fixing the noisy rules it was 89% on a fresh 100.
- **python-logging-study:** static analysis (ast) of logging practice in 21 Python projects; applications log about 8 times more than libraries (2.24 vs 0.28 calls per 1000 lines, p = 0.002).
- **pypi-vulnerability-exposure**: the 300 most downloaded PyPI packages against the OSV database; 33.7% have a known advisory and 29.3% depend directly on one, with only a weak link to dependency count (rho = 0.19).
- **agentic-requirements-engineering**: multi-agent pipeline (Elicitor, Specifier) with defense-in-depth quality gates (pre-tool hooks, AST banned-word scanner, regex quote verification against transcript markers, GitHub Actions CI), benchmarked against an unguided LLM on the MediQueue case study; removed 4 critical failure modes (same-day schedule crash, untestable adjectives, HIPAA front-desk data exposure, 30-minute idle terminal) with 100% linter and CI pass.
- **static-analysis-baseline** (second author, paper in preparation): ran SonarQube Cloud and Codacy on 4 bioinformatics repositories (2,325 Python findings); Codacy reports 18 times more findings per 1000 lines, yet none of 106 rules works at web service level, so neither detects the paper's 9 anti-patterns; TF-IDF clustering was weak (silhouette 0.04 to 0.30) and is reported as descriptive only.

Other projects: parallelised malware detection pipeline; number plate detection with YOLOv5; web and PDF scraping tools.

EXPERIENCE

Associate AI Engineer, Applicon Soft, Lahore | 07/2026 to present

- Set up an LLM-based workflow (Claude) that generates test schemas, database migrations and test suites, and used it to migrate an existing app's backend from Zend to Laravel within a limited token budget.
- Added a speech-enabled chatbot to an existing app (strict instructions, temperature zero) and worked on an agentic retrieval-augmented generation setup with guardrails.
- Built a microphone app that gives pronunciation and accent feedback through a speech API (Mac and Windows).

Freelance content writer and social media manager, remote | 2020 to 07/2026

- Six years of freelance content work: research-based writing for clinicians; multi-platform content production.
- Built a scripted video pipeline: a Python and ExtendScript (After Effects) engine driven by word-level transcripts, with automated captions, brand overlays, subject cut-out, loudness-targeted audio and renders. Claude wrote and iterated the scripts; AI-generated b-roll; engine rebuilt to a client brand specification.

SKILLS

- **Languages:** Python, Java, C++, Dart, JavaScript
- **Backend and infrastructure:** FastAPI, PostgreSQL, Redis, WebSockets, Docker, Firebase
- **Mobile and web:** Flutter (one codebase for web and Android)
- **AI engineering:** LLM prompting, retrieval-augmented generation, agentic workflows, LangChain, guardrails
- **Research and data:** mining software repositories, pandas, SciPy, matplotlib, pytest, git and GitHub
- **Video and social media**: After Effects, Premiere Pro, CapCut, Canva, Meta Ads

CERTIFICATIONS AND LANGUAGES

Cisco Networking Academy: CCNA Introduction to Networks (2024). **Coursera:** Machine Learning (Andrew Ng); IBM Mobile App Development. **Meta:** Digital Marketing Associate (2026).
English: IELTS Academic 7.5 (C1). **Urdu:** native.